→ Numerical

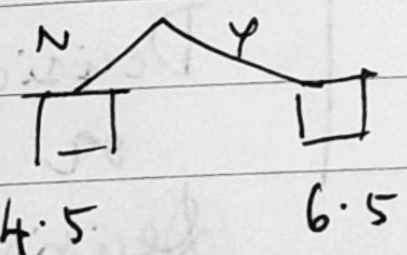
x_1	x_2	y
1	2	4
2	3	5
3	4	6
4	5	7

Iteration 1

a) fit a model f_1

$r_1 = y_1$

$x \geq 2.5$ - Decision Stump



$$r^{(1)} = r^{(1)} - \lambda f_1(x^{(1)})$$

$\lambda = 0.1$

$$= 4 - 0.1(4.5)$$
$$= 3.55$$

$$r^{(2)} = 5 - 0.1(4.5)$$
$$= 4.55$$

$$x^{(3)} = 6 - 0.1(6.5)$$
$$= 5.35$$

$$x^{(4)} = 7 - 0.1(6.5)$$
$$= 6.35$$

$$x = \begin{bmatrix} 3.55 \\ 4.55 \\ 5.35 \\ 6.35 \end{bmatrix}_{4\times 1}$$

for iteration 2

x_1	x_2	x
1	2	3.55
2	3	4.55
3	4	5.35
4	5	6.35

$f_2 \to x$

$x_1 \geq 2.5$

N: $\boxed{4.05}$ Y: $\boxed{5.85}$

$$\frac{3.55 + 4.55}{2}$$

$$x = \begin{bmatrix} 3.145 \\ 4.145 \\ 4.765 \\ 5.765 \end{bmatrix}_{4\times 1}$$

$$x^{(1)} = 3.55 - \lambda f_2(x_0)$$
$$= 3.55 - 0.1(4.05)$$
$$= 3.145$$

$$x^{(2)} = 4.55 - 0.1(4.05)$$
$$= 4.145$$

$$x^{(3)} = 5.35 - 0.1(5.85)$$
$$= 4.765$$

$$x^{(4)} = 5.765$$